

Safety and efficacy of preoperative Bladder Carcinoma Combination-immunotherapy (NABUCCO)

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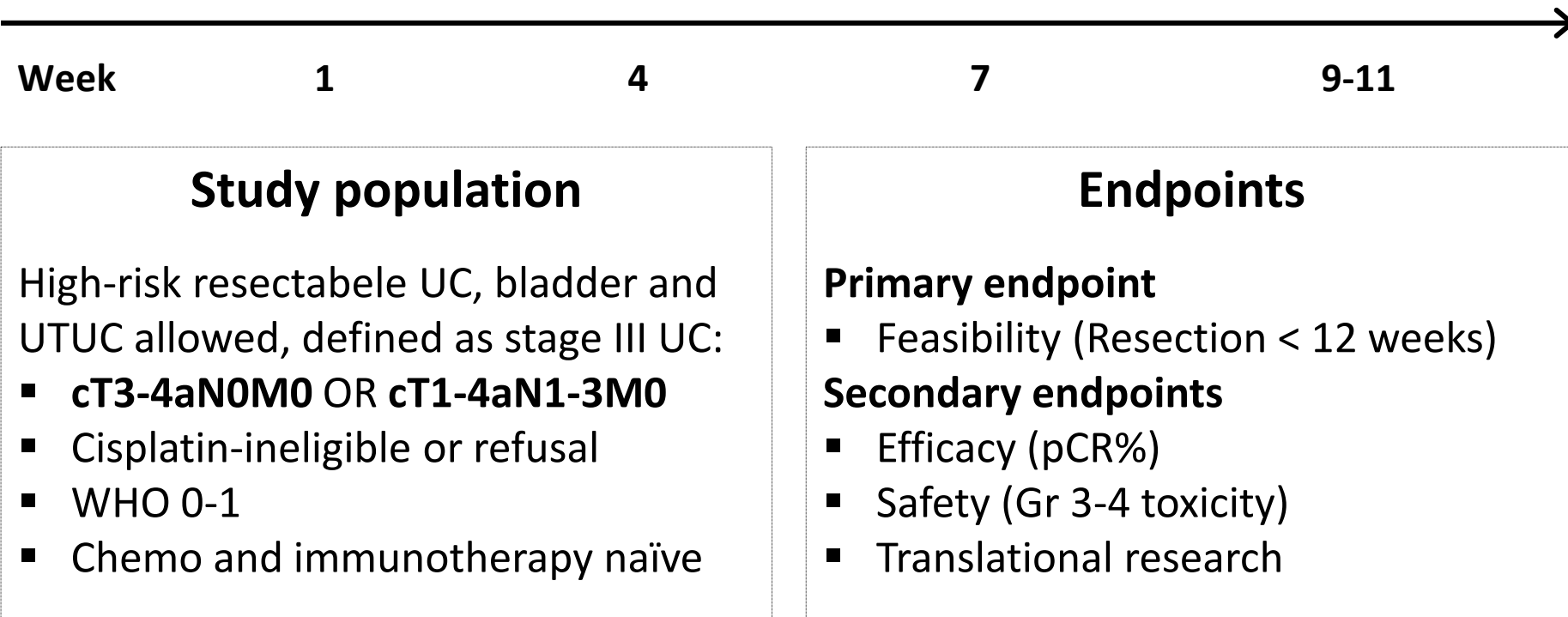
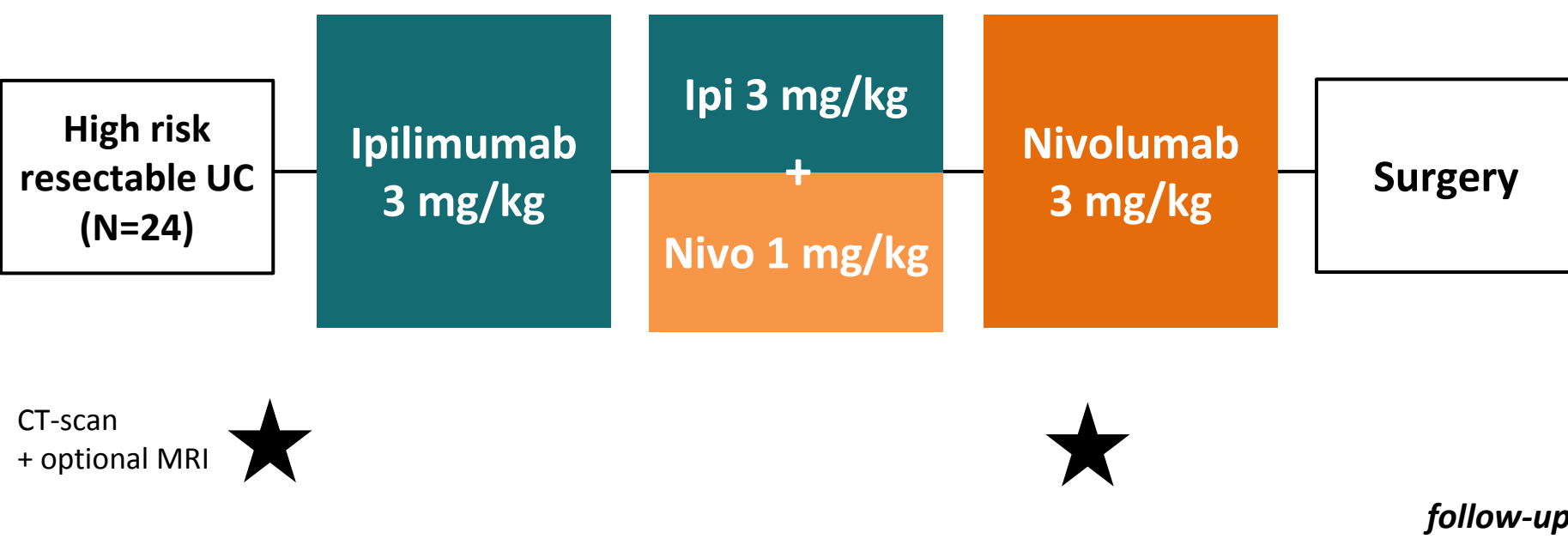
BACKGROUND

- Although muscle-invasive urothelial cancer can be cured by surgery, recurrence rates are high and the 5-year overall survival is only 45-55% for patients with pT3N0 tumors, and even worse for patients having pT4aN0 (35-45%) or pTxN+ (10-35%).
- Overall survival benefit of preoperative cisplatin-based chemotherapy is only 5%, with 30-40% of patients being cisplatin ineligible.
- Immunotherapy targeting the PD-1/PD-L1-axis has shown promising activity and durability in UC, particularly when combined with anti-CTLA-4.
- Encouraging pCR rates (pCR%: 29-42%) were observed in recent clinical trials testing neoadjuvant anti-PD-1/PD-L1 (mostly: cT2-T3N0 bladder cancer pts).
- In this phase 1b single-arm clinical trial, a pre-operative attenuated schedule of ipilimumab and nivolumab was tested in patients with high-risk stage III urothelial cancer patients, as this population with poor prognosis better justifies a more intensive treatment regimen.

References

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STUDY DESIGN



RESULTS

BASELINE CHARACTERISTICS

Characteristics	NABUCCO cohort (N=24)
Median age, years (range)	65 (50-81)
Gender, N(%)	
▪ Male	18 (75%)
▪ Female	6 (25%)
Clinical T stage, N(%)	
▪ WHO 0	21 (88%)
▪ WHO 1	3 (12%)
Clinical T stage, N(%)	
▪ cT3-4aN0M0	14 (58%)
▪ cT1-4aN1-3M0	9 (38%)
▪ cT4bN1M0*	1 (4%)
Location prim tumor	
Bladder	23 (96%)
Bladder + upper tract	1 (4%)
Median PD-L1 CPS, %	32%
PD-L1 positivity, N (%)	
▪ PD-L1 CPS ≥ 10%	15 (63%)
▪ PD-L1 CPS < 10%	9 (37%)

*one patient was staged from T3 to T4b after enrolment, based on MRI and retrospective review of CT-scan

SAFETY AND EFFICACY

Adverse event	All grade (%)	Grade 3/4 (%)
lipase increased	8 (33%)	6 (25%)
fatigue	6 (25%)	1 (4%)
alanine aminotransferase increased	5 (21%)	3 (12%)
diarrhea	5 (21%)	3 (12%)
aspartate aminotransferase increased	4 (17%)	1 (4%)
rash	5 (21%)	0 (0%)
alkaline phosphatase increased	3 (12%)	1 (4%)
colitis	3 (12%)	2 (8%)
ggt increased	3 (12%)	2 (8%)
headache	3 (12%)	0 (0%)
hyperthyroidism	3 (12%)	0 (0%)
serum amylase increased	3 (12%)	0 (0%)
adrenal insufficiency	2 (8%)	0 (0%)
hypothyroidism	2 (8%)	0 (0%)
toxicoderma, DRESS syndrome	1 (4%)	1 (4%)
blood bilirubin increased	1 (4%)	1 (4%)
Peripheral motor neuropathy	1 (4%)	1 (4%)
hemolysis	1 (4%)	1 (4%)
hyperglycemia	1 (4%)	1 (4%)

Safety	All pts (n=24)
Surgical resection, N(%)	24 (100%)
Surgical resection < 12 weeks (primary endpoint)	23 (96%)
Treatment cycles, N(%)	
▪ pts having 3 treatment cycles	18 (75%)
▪ pts having 2 treatment cycles	6 (25%)
Treatment-related AEs, N(%)	
▪ Grade 3/4	13 (55%)
▪ Grade 3/4 (when excluding elevated clinically insignificant lipase)	10 (41%)
30-day mortality rate, N(%)	0 (0%)
90-day mortality rate, N(%)	1 (4%), widespread metastases

Pathological response	All pts(n=24)
pCR (secondary endpoint)	11 (46%)
Noninvasive disease (11 pCR, 2 CIS only, 1 pTa)	14 (58%)
pCR per subgroup	
▪ PD-L1 CPS ≥ 10% (n=15)	9 (60%)
▪ PD-L1 CPS < 10% (n=9)	2 (22%)
Noninvasive disease per subgroup	
▪ PD-L1 CPS ≥ 10% (n=15)	11 (73%)
▪ PD-L1 CPS < 10% (n=9)	3 (33%)
pCR per LN status	
▪ cN0 (n=14)	7 (50%)
▪ cN+ (n=10)	4 (40%)
pCR per treatment group	
▪ pts having 2 cycles (n=6)	1 (17%)
▪ pts having 3 cycles (n=18)	10 (56%)

EFFICACY

- No patients had on-treatment radiologic or clinical progression.
- High rates of pCR (46%) and noninvasive disease (58%) were observed.
- 2 additional patients had near pCR (<10% residual vital tumor, pN0).
- 1 patient recurred 6 weeks after surgery, and died of metastatic disease.
- Responses were numerically higher in PD-L1+ pts.

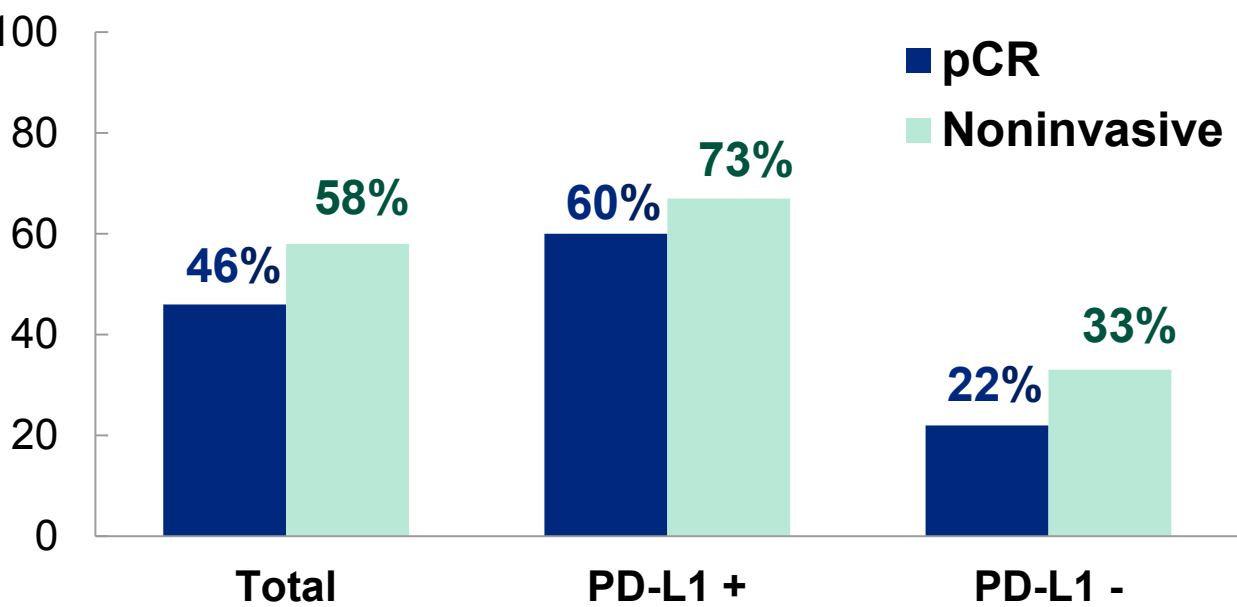


Figure 1. pCR and noninvasive disease (ypCR, ypTisN0, ypTaNo) rate by PDL1 CPS score. PDL1+ is defined as CPS>10.

MULTIPLEX IMMUNOFLOURESCENCE RESPONSE PATTERNS

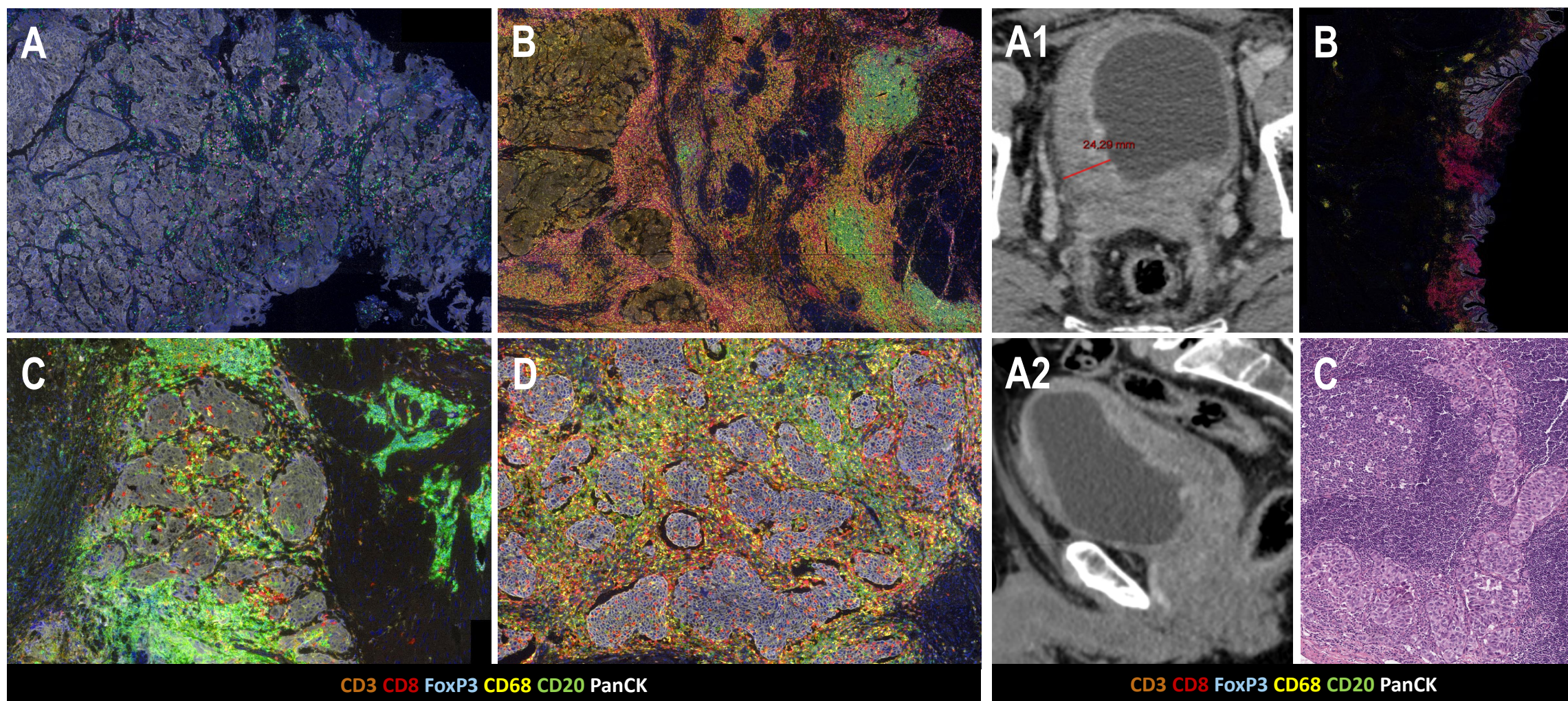


Figure 2. Multiplex immunofluorescence (mIF) imaging of baseline TUR and cystectomy tissue of MIBC patients treated with pre-operative ipilimumab/nivolumab. (A) TUR sample of a patient with cT3N0 MIBC showing absence of CD8 T cells. (B) After 3 treatment cycles, a complete response (pCR) by pathological examination, showing fields of necrosis (pan-CK still present but no nuclei) and abundant immune infiltrate. (C) TUR of cT4aN1 MIBC patient with CD8 T cell infiltration and abundant CD20 infiltration. (D) Cystectomy showed a near pCR (<10% residual viable tumor in bladder, ypT2N0), revealing small remaining tumor nests that are highly infiltrated by CD8-positive T-cells, surrounded by CD20 and CD68 positive cells.

Figure 3. A 66 year old patient that was diagnosed with cT4bN1 MIBC (A1, A2) after being upstaged after enrolment based on MRI. Pathological examination showed a pCR in the primary tumor with CD8 T cell infiltrates in the scar tissue (B). Reactive fibrosis was observed in several LNs, and 1 LN showed a micro-metastasis with vital tumor showing no response or immune cell infiltration (C) – ypT0N1.

FEASIBILITY

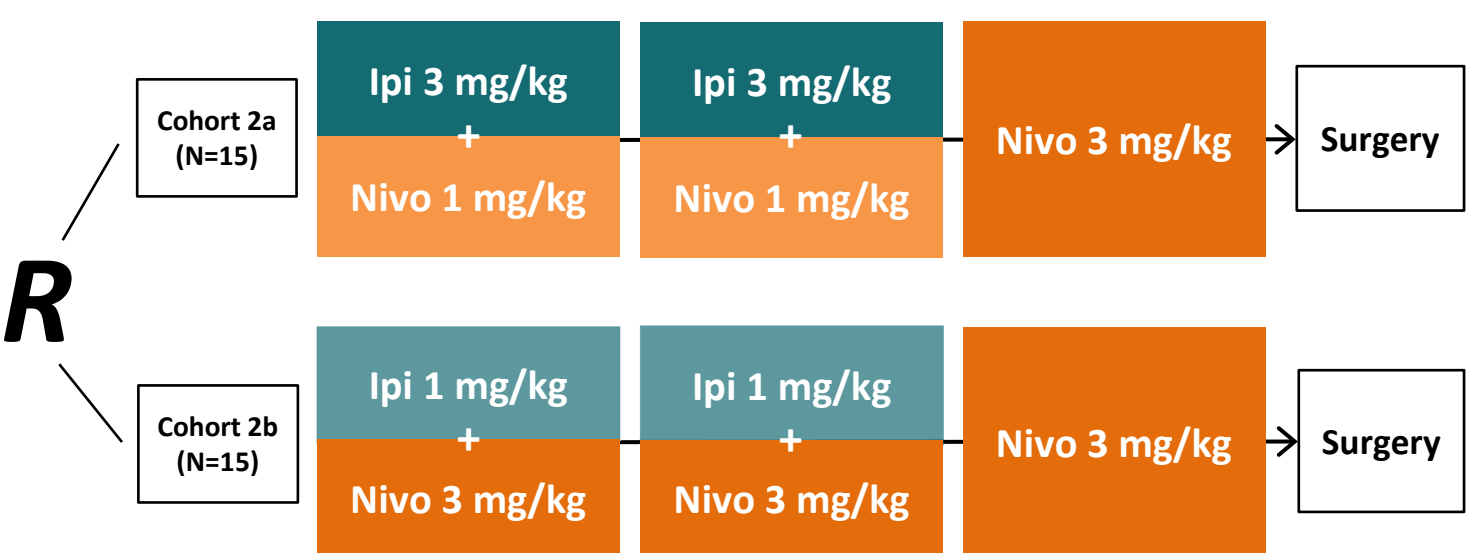
Pre-operative ipi/nivo did not endanger resection in the current setting.

- All patients had resection of their tumor.
- 23 (96%) patients had surgical resection within 12 weeks from first infusion, and 1 (4%) patient had a 4 week delay due to a TRAE (hemolysis).
- 18 (75%) of patients received all 3 treatment cycles, and 6 (25%) received only 2 cycles due to TRAEs.

CONCLUSIONS

- All patients had surgical resection; 96% within 12 weeks
- Robust response rates (pCR 46%; 58% noninvasive disease) were observed with the current treatment schedule in locoregionally advanced UC.
- Additional patients had major pathological responses, the clinical significance of these responses is unknown yet.

Follow-up cohorts: high vs low ipilimumab and frontline nivolumab



Contact information

This poster was presented at the European Society of Medical Oncology Annual Meeting (Sept 27th – Oct 1st, 2019): Barcelona, Spain
No conflicts of interest to declare by presenting author
NABUCCO is sponsored by Bristol-Myers Squibb
Clinical Trial identification: NCT03387761.

Abstract #2053
FPN: 904PD